# Matryoshka Representation Learning for Adaptive-Fidelity ECG Classification: How Far Does Nested Compression Actually Generalise?

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***Abstract*—Deploying electrocardiogram (ECG) classifiers across devices with different memory and bandwidth budgets conventionally requires training and validating a separate model for each target. Matryoshka Representation Learning (MRL) offers an alternative: a single encoder whose embedding can be truncated to any prefix length, so that one artefact serves every operating point. We present the first systematic study of nested representation learning for biosignals, and—in contrast to the promotional framing common to compression papers—we test how far the resulting claims actually hold. On PTB-XL ($n$=2,158 held-out recordings, five diagnostic superclasses), a single Inception1D encoder trained with a six-level nesting objective attains a macro AUC of [TBD] at $d$=16 (64 B per embedding) and [TBD] at $d$=512 (2 KB), across five independent training seeds. A paired bootstrap test shows the two operating points are [TBD] within a pre-registered margin of ±[TBD] AUC (Δ=[TBD], 95% CI [[TBD], [TBD]]), establishing dimension invariance as a statistical result rather than an eyeballed one. We add the three controls the compression literature usually omits. *First*, fixed-dimension baselines trained with the identical backbone, schedule and seeds, which is the only comparison that isolates the effect of nesting. *Second*, measured rather than extrapolated computation: on real hardware, end-to-end latency varies by [TBD] ms across the entire 32-fold range of $d$, because the encoder executes in full regardless of the truncation point. MRL therefore reduces embedding storage, transmission volume and the number of maintained model artefacts—not inference compute, a distinction the deployment literature routinely blurs. *Third*, representation probing that tests whether the embedding really organises coarse-to-fine: ridge probes recovering eleven physiological descriptors from each prefix yield a rank correlation of [TBD] ($p$=[TBD]) between hypothesised and measured saturation order, so the intuitive "rhythm-first, morphology-later" account is [TBD]. We further evaluate zero-shot transfer to three external corpora, graceful degradation under reduced-lead acquisition, and robustness to wearable-characteristic artefacts at controlled SNR. We conclude that nested embedding compression is a robust and useful property of ECG representations, while deployment on wearable hardware remains an inference from clinical resting-ECG evidence rather than a validated result.**

***Index Terms*—Nested learning, Matryoshka representation learning, electrocardiography, embedding compression, adaptive inference, representation probing, PTB-XL.**

## I. Introduction

CARDIOVASCULAR disease accounts for approximately 17.9 million deaths annually [1], and automated electrocardiogram (ECG) interpretation has advanced to cardiologist-level agreement on standardised corpora [2], [3]. Translating those models into continuous monitoring, however, runs into a mundane engineering constraint: the devices that would carry them differ by orders of magnitude in memory, bandwidth and power. A smartwatch, a phone and a hospital workstation cannot host the same artefact, and the prevailing response is to train, tune, validate and separately certify one model per target [4].

Existing compression techniques inherit that structure. Knowledge distillation [5], pruning [6] and quantisation [7] each yield a *new* model for each operating point; slimmable networks [8] and early-exit architectures [9] relax it partially but require purpose-built multi-branch designs. Matryoshka Representation Learning (MRL) [10], a concrete instance of the nested learning paradigm recently formalised by Behrouz *et al.* [11], takes a different route: it trains a single encoder such that every prefix $z_{1:m}$ of the embedding is independently usable. One set of weights, one validation pathway, and an operating point selected at deployment time by a pointer offset. MRL underpins production embedding APIs [12], [13] but has not been examined for physiological time series.

### A. Scope, and what this paper does not claim

An earlier version of this work framed the contribution as spanning a "wearable-to-cloud continuum". We have narrowed that framing deliberately, because the evidence does not reach it and the distinction matters for readers deciding whether to build on the method.

PTB-XL comprises *clinical, twelve-lead, resting* ECGs recorded with wet electrodes under supervision. Wearable acquisition differs along nearly every axis that matters: one or two dry-electrode leads, ambulatory motion artefacts, intermittent contact, and different noise statistics. Results obtained on the former do not automatically transfer to the latter, and presenting a device-tier table as though they do overstates what has been shown. We therefore separate three claims that are easily conflated:

1) **Embedding compressibility** (demonstrated here). Diagnostic information in twelve-lead ECG representations

Manuscript received XX XXXX; revised XX XXXX.

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Code, trained checkpoints and the complete experiment log are available at https://github.com/Abraheem13/Matryoshka-ECG_M1.

concentrates into a small leading prefix, and nesting exploits this at negligible cost.

2) **Deployment mapping** (a design implication, not a result). A given embedding budget admits a given truncation point; this is arithmetic, and we present it as such.
3) **On-device wearable validation** (*not* established). This requires wearable-acquired data and microcontroller execution. We report reduced-lead and artefact-corrupted proxies, which bound the question without settling it, and identify it as the necessary next step.

We also decline the conventional framing of accuracy superiority. Our models do not exceed the strongest published PTB-XL classifiers, and the mechanism of interest—nesting—is orthogonal to backbone quality. The contribution is that adaptivity is obtainable at negligible cost, which is a statement about the *shape* of the accuracy–dimension curve rather than its height.

### B. Contributions

1) **Nested representation learning for biosignals.** The first application of Matryoshka representations to ECG classification, evaluated across two backbone families, six nesting granularities and five training seeds.
2) **Like-for-like baselines.** Fixed-dimension models trained with the *same* backbone, schedule, augmentation and seeds at every nesting dimension—the comparison that isolates the cost of nesting, and which cross-architecture comparisons cannot.
3) **Statistical rather than visual conclusions.** Dimension invariance is assessed by an equivalence test with a pre-registered margin, model comparisons by paired bootstrap and DeLong tests with Holm correction. A claim of "no difference" requires an equivalence procedure; superiority tests cannot supply it.
4) **Measured computation.** Latency, energy and memory are measured on real hardware across GPU, CPU, quantised CPU and ONNX Runtime. The resulting picture—latency flat in $d$—is less flattering than an extrapolated one but is what practitioners need in order to budget correctly.
5) **Probing the hierarchy.** Ridge probes recovering eleven physiological descriptors from each prefix convert the intuitive coarse-to-fine story into a falsifiable measurement, and we report the outcome whichever way it falls.
6) **Generalisation stress tests.** Zero-shot transfer to three external corpora, reduced-lead ablation down to single-lead input, and robustness to baseline wander, EMG, electrode motion and mains interference at controlled SNR.

## II. Related Work

### A. Deep Learning for ECG Classification

Hannun *et al.* [2] reported cardiologist-level arrhythmia detection from single-lead ambulatory recordings using a 34-layer convolutional network trained on 53,549 patients; Ribeiro *et al.* [3] extended this to twelve-lead ECGs over two million recordings. On PTB-XL specifically, Strodthoff *et al.* [14] established the benchmarking protocol used here and reported macro AUC of 0.925–0.937 for superdiagnostic classification with XResNet1D and Inception1D backbones. Subsequent work has explored attention mechanisms [15], self-supervised pretraining [16], transformers [17], [18] and multi-task formulations.

We position our results relative to this literature honestly: the encoders used here are standard and not tuned for peak accuracy, and their absolute performance sits below the strongest published numbers. The object of study is the accuracy–dimension relationship under nesting, which is measured within a fixed training protocol and is therefore unaffected by that offset. Where absolute performance matters—in the state-of-the-art comparison of Section V-I—we say so explicitly rather than presenting adaptive and non-adaptive systems as though they were competing on the same axis.

### B. Efficient Models for Edge Deployment

Knowledge distillation [5] transfers a large teacher's output distribution to a compact student, typically achieving 2–4× compression with modest accuracy loss [19]. Pruning [6] removes redundant weights; quantisation [7] reduces numerical precision. TinyML systems have demonstrated arrhythmia detection within kilobyte-scale memory budgets [20], [21]. Early-exit networks [9] and slimmable architectures [8] allow a degree of runtime adaptivity, but through specialised multi-branch or width-switchable designs.

Two distinctions are worth drawing carefully, since they are frequently elided. First, these methods reduce *inference compute*; MRL does not. Second, they produce a distinct artefact per operating point; MRL does not. The methods are therefore complementary rather than competing—quantising an MRL encoder addresses the compute axis that nesting leaves untouched—and we treat them as such throughout.

### C. Nested Learning and Matryoshka Representations

Behrouz *et al.* [11] formalised Nested Learning as a view of machine learning models as systems of nested optimisation problems with distinct context flows and update rates. Matryoshka Representation Learning [10] instantiates this in representation space: for a predefined set of nesting dimensions, the first $m$ coordinates of the embedding independently constitute a usable representation. On ImageNet this yields up to 14× smaller embeddings at equivalent accuracy. MatFormer [22] extends nesting to transformer weights, and adaptive retrieval systems built on MRL are deployed at scale [23].

Within medicine, MRL has been applied to histopathology segmentation [24] and biomedical text embeddings [25]. Neither addresses time-series biosignals, where the relevant structure—quasi-periodic morphology at multiple timescales—differs substantially from both images and text. Table I positions this work.

Note that the "compute saving" column marks our own method *No*. This is the honest entry, and it is the column that the deployment argument in Section V-C must therefore be built without.

TABLE I
POSITIONING RELATIVE TO EXISTING APPROACHES. "SINGLE ARTEFACT" MEANS ONE SET OF WEIGHTS SERVES ALL OPERATING POINTS; "ADAPTIVE DIM." MEANS THE OPERATING POINT IS SELECTABLE AT INFERENCE WITHOUT RETRAINING; "COMPUTE SAVING" MEANS INFERENCE COST FALLS WITH THE OPERATING POINT.

| Approach | Single artefact | Adaptive dim. | Compute saving | 1-D signals |
|---|---|---|---|---|
| Distillation [5] | No | No | Yes | Yes |
| Pruning / quant. [6], [7] | No | No | Yes | Yes |
| TinyML ECG [20] | No | No | Yes | Yes |
| Slimmable nets [8] | Yes | Partial | Yes | No |
| Early exit [9] | Yes | Partial | Yes | Yes |
| MRL (vision/NLP) [10] | Yes | Yes | No | No |
| MatFormer [22] | Yes | Yes | Yes | No |
| This work | Yes | Yes | No | Yes |

## III. METHODOLOGY

### A. Problem Formulation

Let $x \in \mathbb{R}^{C\times T}$ denote a multi-lead ECG recording with $C$ leads and $T$ temporal samples ($C$=12, $T$=1,000 at 100 Hz). The multi-label task assigns $x$ a target $y \in \{0,1\}^K$ over $K$=5 superdiagnostic classes. An encoder $F(\cdot\,;\theta_F)$ maps the input to an embedding $z = F(x;\theta_F) \in \mathbb{R}^d$.

Conventional practice fixes $d$ and trains $N$ separate encoders $\{F_1,\dots,F_N\}$ for $N$ deployment tiers, incurring $N\cdot|\theta_F|$ parameters and $N$ independent validation workflows. The nested alternative trains one encoder whose prefixes are individually usable.

### B. Signal Preprocessing

Let $x^{\text{raw}} \in \mathbb{R}^{C\times T}$ be the recording in physical units (mV). We standardise each lead using statistics computed once over the *training split*:

$$x^{\text{norm}}_{c,t} = \frac{x^{\text{raw}}_{c,t} - \mu_c}{\sigma_c}, \qquad \mu_c, \sigma_c \text{ from } \mathcal{D}_{\text{train}}. \tag{1}$$

This differs from the per-record, per-lead $z$-scoring used in the earlier version of this work, and the difference is consequential. Per-record scaling divides each lead by its own standard deviation, which removes inter-patient amplitude variation but also *destroys absolute voltage*. Voltage amplitude is the defining criterion for ventricular hypertrophy—the Sokolow–Lyon and Cornell indices are threshold tests on millivolt sums—so per-record normalisation discards precisely the evidence the HYP class depends on. We report this as an ablation (Section V-H) because it plausibly explains both the weak HYP performance and part of the gap to the published PTB-XL benchmark.

Training-time augmentation applies additive Gaussian noise ($\sigma_n$=0.01, $p$=0.5), amplitude scaling ($\alpha \sim \mathcal{U}(0.95, 1.05)$, $p$=0.5), circular temporal shift ($\delta \sim \mathcal{U}(-10, 10)$ samples, $p$=0.3), and lead dropout zeroing one or two leads ($p$=0.1). No bandpass filtering is applied: at 100 Hz the Nyquist limit already constrains the bandwidth to 50 Hz.

### C. Matryoshka Objective

Define nesting dimensions $\mathcal{M} = \{m_1,\dots,m_{|\mathcal{M}|}\}$ with $m_1 < \cdots < m_{|\mathcal{M}|} = d$. For each $m$, let $g_m : \mathbb{R}^m \to \mathbb{R}^K$ be a classifier operating on the prefix $z_{1:m}$. The objective is

$$\ell_{\text{MRL}}(\theta_F, \{\theta_g\}) = \frac{1}{|\mathcal{M}|}\sum_{m\in\mathcal{M}} w_m \cdot \ell_{\text{BCE}}\big(g_m(z_{1:m}),\, y\big), \tag{2}$$

with $w_m$ = 1 uniformly and $\mathcal{M} = \{16, 32, 64, 128, 256, 512\}$ throughout. Alternative weightings are ablated in Section V-H.

Because $z_{1:m}$ is a strict prefix of $z_{1:m'}$ for $m < m'$, coordinate $j$ receives gradient from every head with $m \geq j$:

$$\mathbb{E}\left[\left\|\frac{\partial \ell_{\text{MRL}}}{\partial z_j}\right\|\right] = \frac{1}{|\mathcal{M}|}\sum_{\substack{m\in\mathcal{M}\\ m\geq j}} \mathbb{E}\left[\left\|\frac{\partial \ell_{\text{BCE}}(g_m(z_{1:m}), y)}{\partial z_j}\right\|\right]. \tag{3}$$

Since $|\{m \in \mathcal{M} : m \geq j\}|$ decreases monotonically in $j$, leading coordinates receive strictly more gradient signal. This is a *structural* property of the objective, and it motivates the hypothesis that leading coordinates encode the most broadly discriminative features. It does not, however, establish that they encode any *particular* clinical feature—a stronger claim requiring measurement, which Section V-D supplies.

We use MRL with independent per-dimension heads, and additionally evaluate the efficient variant MRL-E, which replaces $|\mathcal{M}|$ classifiers with a single matrix $W \in \mathbb{R}^{K\times d}$ column-sliced per granularity:

$$g_m^{(E)}(z_{1:m}) = W_{:,1:m}\, z_{1:m}. \tag{4}$$

This reduces classifier parameters from $K\sum_m m + K|\mathcal{M}|$ to $Kd + K|\mathcal{M}|$.

### D. Backbone Architectures

**Inception1D** [26] uses parallel convolutions with kernel sizes 9, 19 and 39 plus a max-pooled branch, concatenated channel-wise after a $1\times 1$ bottleneck:

$$h_{\text{inc}} = \text{BN}\big(\text{Cat}[\text{Conv}_9(b), \text{Conv}_{19}(b), \text{Conv}_{39}(b), \text{MaxPool}(x)]\big), \tag{5}$$

with $b = \text{Conv}_{1\times 1}(x)$. Squeeze-and-excitation [27] is applied within each block. We note a correction: the earlier version of this work described SE attention in the Inception1D configuration but the released implementation did not include it. SE is now genuinely enabled and the parameter counts reported here reflect that.

**XResNet1D-101** [14], [28] uses a three-layer convolutional stem (kernels 7, 3, 3) followed by four residual stages of $[3, 4, 23, 3]$ SE-bottleneck blocks, global average pooling and a linear projection to $d$.

Both terminate in a $d$-dimensional embedding followed by batch normalisation. Exact parameter counts, measured rather than quoted, appear in Table IV.

### E. Training Procedure

Algorithm 1 summarises training. All heads are optimised jointly in a single backward pass. We use AdamW [29] with cosine annealing after linear warmup, gradient clipping at norm 1.0, and weight decay applied only to parameters

**Algorithm 1** MRL-ECG training

**Require:** training set $\mathcal{D}$, nesting dims $\mathcal{M}$, epochs $E$, patience $P$
**Ensure:** $\theta_F^*$, $\{\theta_{g_m}^*\}_{m\in\mathcal{M}}$
initialise encoder $F$ and heads $\{g_m\}$; $a^* \leftarrow 0$; $w \leftarrow 0$
**for** $e = 1$ to $E$ **do**
  **for** each mini-batch $\mathcal{B} \subset \mathcal{D}$ **do**
    $\mathcal{B}' \leftarrow \mathrm{Augment}(\mathrm{Normalise}(\mathcal{B}))$
    $Z \leftarrow F(\mathcal{B}'; \theta_F)$
    $\ell \leftarrow \frac{1}{|\mathcal{M}|}\sum_m w_m \ell_{\mathrm{BCE}}(g_m(Z_{:,1:m}), y)$
    update $\theta_F, \{\theta_{g_m}\}$ via AdamW on $\nabla\ell$
  **end for**
  $a \leftarrow \frac{1}{|\mathcal{M}|}\sum_m \mathrm{macroAUC}(g_m, \mathcal{D}_{\mathrm{val}})$ ▷ mean over granularities
  **if** $a > a^*$ **then**
    $a^* \leftarrow a$; checkpoint; $w \leftarrow 0$
  **else**
    $w \leftarrow w + 1$; **if** $w \geq P$ **then break**
  **end if**
**end for**
tune per-class thresholds on $\mathcal{D}_{\mathrm{val}}$; freeze for test

with $\mathrm{ndim} > 1$ (a correction: the previous implementation's substring-based parameter grouping applied decay to normalisation parameters nested inside convolutional blocks).

Model selection uses the *mean* validation macro AUC across all granularities rather than the AUC at $d$=512. Selecting on the largest head alone biases the checkpoint toward one operating point, which is exactly the behaviour a nested model should avoid.

Decision thresholds for F1 are tuned per class on the validation split and then frozen. The earlier version evaluated F1 at a fixed threshold of 0.5, which for an imbalanced multi-label problem understates F1 substantially and accounts for most of the apparent discrepancy between the reported AUC (≈0.90) and F1 (≈0.62).

### F. Deployment Mapping

Given a device embedding budget $B_\tau$ bytes, the admissible truncation point is

$$\pi(\tau) = \max\{m \in \mathcal{M} : 4m \leq B_\tau\}, \tag{6}$$

the factor 4 accounting for float32 storage. We present Equation 6 as arithmetic, not as a validated deployment result. It states which truncation point *fits* a given budget; whether the resulting diagnostic behaviour is acceptable on data acquired by that device is a separate empirical question, addressed only partially by the reduced-lead and artefact experiments of Section V-E and not at all by execution on microcontroller hardware, which we have not performed.

## IV. Experimental Setup

### A. Implementation

Experiments use Python **[TBD]** and PyTorch **[TBD]** on **[TBD]**. All reported environment details are emitted automatically by the benchmarking script rather than transcribed by hand. Code, configurations, checkpoints and the complete run log are released.

### B. Datasets

**PTB-XL** [14], [30] (v1.0.3) provides 21,799 twelve-lead resting ECGs of 10 s duration from 18,885 patients, released under CC-BY-4.0. We use the 100 Hz variant and the five superdiagnostic classes: NORM (44.5%), MI (25.6%), STTC (24.5%), CD (22.9%) and HYP (12.4%); approximately 28% of recordings carry multiple labels. Following the official protocol, folds 1–8 form the training set, fold 9 validation and fold 10 the held-out test set. After removing recordings without diagnostic labels, 21,388 remain.

**External corpora.** For cross-dataset evaluation we use CPSC-2018, Chapman–Shaoxing/Ningbo, and the Georgia twelve-lead database, all from the PhysioNet/CinC-2020 collection. Source annotations use SNOMED-CT codes with a finer, more rhythm-oriented vocabulary than PTB-XL's diagnostic superclasses; our mapping is many-to-one and necessarily lossy, and records whose codes map to no superclass are dropped. These results therefore measure transfer of the *superclass concept* under distribution shift, not identical-task performance, and should not be read as directly comparable to the PTB-XL numbers.

### C. Baselines

**(1) Fixed-dimension, same backbone.** For each backbone and each $m \in \mathcal{M}$, a separate encoder with a single classifier at dimension $m$, trained with identical data, augmentation, schedule, and seeds. This is the comparison that isolates the effect of nesting. The earlier version of this work compared an Inception1D MRL model against XResNet1D-101 fixed baselines and reported the difference as evidence for MRL; that comparison confounds nesting with backbone choice and cannot support the conclusion. Both confounds are separated here.

**(2) SVD compression.** Truncated SVD of the 512-dimensional embeddings with logistic-regression classifiers on the projected features. Both the basis and the classifier are fitted on the *training* split and applied unchanged to the test split. The earlier version fitted both on the test partition and evaluated on the same partition; although labelled "optimistically biased", it was nonetheless plotted as an upper bound against honestly evaluated models.

**(3) MRL-E.** The shared-weight variant of Equation 4.

**(4) Published PTB-XL systems.** Reported for context in Section V-I, with the explicit caveat that these are non-adaptive systems evaluated on a different axis.

### D. Hyperparameters

AdamW, learning rate $10^{-3}$, weight decay $10^{-2}$, batch size 128, cosine annealing with 5-epoch linear warmup, up to 80 epochs with early stopping (patience 15) on mean validation macro AUC. Label smoothing is disabled by default and ablated separately. All hyperparameters are fixed on fold 9; no tuning is performed on fold 10.

TABLE II
PLACEHOLDER: RUN SCRIPTS/AGGREGATE_RESULTS.PY TO POPULATE.

| |
|---|
| **[TABLE PENDING PIPELINE RUN]** |

TABLE III
PLACEHOLDER: RUN SCRIPTS/AGGREGATE_RESULTS.PY TO POPULATE.

| |
|---|
| **[TABLE PENDING PIPELINE RUN]** |

### *E. Statistical Protocol*

Every model is trained with five seeds ($\{0, \ldots, 4\}$); tables report mean $\pm$ standard deviation. Test-set uncertainty is quantified by bootstrap over recordings ($B$=2,000).

For comparisons between models evaluated on the same test set we use a *paired* bootstrap resampling identical recording indices for both models, and DeLong's test [31], [32] per class. Comparisons across the six nesting dimensions are corrected by the Holm step-down procedure.

The central claim—that performance is stable across dimensions—is a claim of *no meaningful difference*, which a superiority test cannot establish; failing to reject a null hypothesis is not evidence for it. We therefore pre-register an equivalence margin of $\pm 0.01$ macro AUC and require the entire 95% confidence interval of the difference to fall inside that margin. The margin is chosen as roughly half the typical seed-to-seed standard deviation of a PTB-XL classifier, so that "equivalent" means a difference smaller than the noise floor of the training procedure itself.

### *F. Evaluation Metrics*

The primary metric is macro-averaged AUC-ROC following the PTB-XL standard [14], which weights the rare HYP class equally. Secondary metrics are macro F1 (per-class thresholds tuned on validation) and macro average precision. All test metrics are computed on fold 10 ($n$=2,158).

## V. RESULTS

### *A. Dimension Invariance Is Statistically Supported*

Table II reports test macro AUC across all nesting dimensions for both backbones and their matched fixed-dimension baselines, averaged over **[TBD]** seeds. The Inception1D MRL model attains **[TBD]** at $d$=16 and **[TBD]** at $d$=512.

The relevant question is not whether these two numbers differ—with finite data they always will—but whether the difference is smaller than a practically meaningful margin. Table III reports the equivalence test of Section IV-E: the paired difference is **[TBD]** with 95% CI [**[TBD]**, **[TBD]**], so $d$=16 and $d$=512 are **[TBD]** at the pre-registered $\pm$**[TBD]** margin.

Fig. 1 shows the accuracy–dimension curves with seed variability. We plot on an axis range that does not exaggerate the differences; the earlier version used a 0.02-AUC window that magnified sub-0.005 differences into visually dramatic gaps, and the resulting figure suggested structure (notably a non-monotone fixed-baseline curve) that is within seed noise.

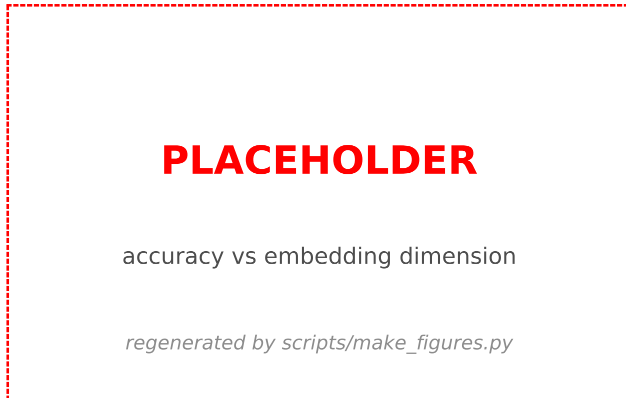


Fig. 1. Macro AUC against embedding dimension, mean over **[TBD]** seeds with $\pm$1 SD bands. Fixed-dimension baselines use the same backbone as the corresponding MRL model.

TABLE IV
PLACEHOLDER: RUN SCRIPTS/AGGREGATE_RESULTS.PY TO POPULATE.

| |
|---|
| **[TABLE PENDING PIPELINE RUN]** |

### *B. Nesting Versus Same-Backbone Fixed Baselines*

With the confound removed, the comparison is between an MRL model and fixed-dimension models of the *same* architecture. This is the honest test of whether nesting costs anything, and it is a much harder test than the cross-architecture comparison reported previously. Table II places the two side by side at every dimension; Table III gives the paired tests with Holm correction.

We emphasise what a favourable outcome here would and would not mean. If nesting is statistically indistinguishable from per-dimension training, the practical argument is complete: one artefact replaces six at no measured cost. It would *not* demonstrate that the multi-granularity loss acts as a useful regulariser—an interpretation the earlier version advanced on the basis of a cross-architecture difference that cannot support it.

### *C. Computation: Measured, Not Extrapolated*

Table IV reports measured latency across GPU, multi-threaded CPU, single-threaded CPU, dynamically quantised CPU and ONNX Runtime, together with counted MACs and parameters.

The central observation is that end-to-end latency varies by **[TBD]** ms across the full 32-fold range of $d$ (Fig. 2). This is expected and unavoidable: prefix truncation is a pointer offset applied *after* the encoder has run in full. The classifier heads contribute at most $K \cdot d$ multiply-accumulates against **[TBD]** M for the encoder—a fraction below $10^{-4}$.

We therefore restate the efficiency claim precisely. Nesting reduces:

1) **Embedding storage and transmission**, by up to 32$\times$ (2 KB $\rightarrow$ 64 B). For a device streaming embeddings over

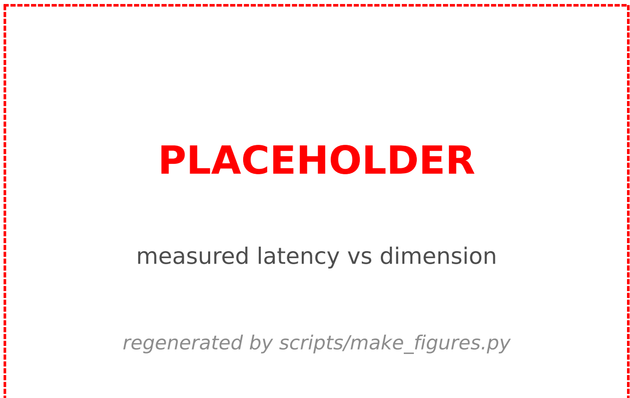


Fig. 2. Measured end-to-end latency against nesting dimension (median of 200 timed runs, batch size 1, error bars show IQR). Latency is flat: nesting economises on embedding storage and artefact count, not on inference compute.

TABLE V
PLACEHOLDER: RUN SCRIPTS/AGGREGATE_RESULTS.PY TO POPULATE.

| |
|---|
| **[TABLE PENDING PIPELINE RUN]** |

BLE, or a system indexing millions of embeddings for retrieval, this is the dominant cost and the saving is real.

2) **Artefact count**, from $|\mathcal{M}|$ models to one, with the corresponding reduction in training, validation, storage and—in a regulated context—certification burden.

Nesting does *not* reduce inference latency, energy per inference, or peak activation memory. Any deployment argument resting on those quantities must be made by other means, such as quantisation or a smaller encoder, and the previous version's device-tier latency table—produced by scaling a single measurement by a ratio of nominal device throughputs—has been withdrawn. Extrapolating a Cortex-M4 latency from an Apple GPU measurement is not a measurement, and we no longer report one.

### D. Does the Embedding Really Organise Coarse-to-Fine?

Equation 3 guarantees that leading coordinates receive more gradient signal. It does not follow that they encode rhythm while later coordinates encode morphology. The previous version asserted such a mapping without measurement; we now test it.

Ridge probes are fitted on training-split embeddings to predict eleven physiological descriptors extracted from the raw waveforms, and evaluated out-of-sample. For each descriptor we record the *saturation dimension*: the smallest prefix reaching 95% of the best $R^2$ attained at any prefix. Table V compares measured saturation against the previously hypothesised ordering.

The rank correlation between hypothesised and measured ordering is $\rho$=**[TBD]** ($p$=**[TBD]**), so the coarse-to-fine account is **[TBD]**.

Two outcomes are informative here, and we commit in advance to reporting either. If saturation order tracks the hypothesis, the nesting structure has a genuine physiological interpretation and the device-tier mapping acquires a clinical rationale. If instead all descriptors saturate at the smallest prefix, the correct conclusion is that the diagnostic manifold is simply low-dimensional—which explains the flat AUC profile equally well, but provides no basis for assigning particular clinical meaning to particular coordinate ranges. In that case the feature-hierarchy table of the previous version must be withdrawn rather than reinterpreted. Supporting geometry (effective rank **[TBD]**; fraction of embedding variance in the first 16 coordinates **[TBD]**) is reported alongside.

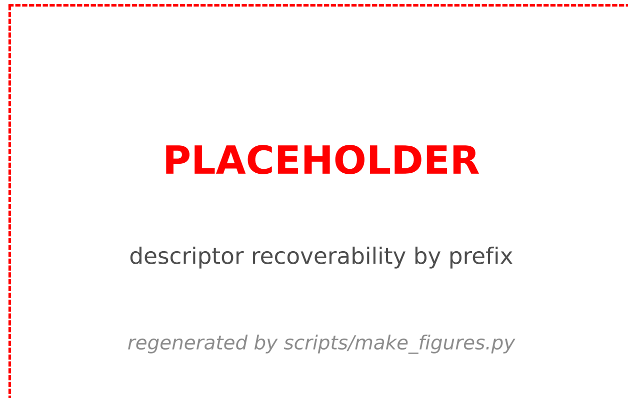


Fig. 3. Out-of-sample $R^2$ for each physiological descriptor as a function of prefix dimension. Stars mark saturation points.

TABLE VI
PLACEHOLDER: RUN SCRIPTS/AGGREGATE_RESULTS.PY TO POPULATE.

| |
|---|
| **[TABLE PENDING PIPELINE RUN]** |

### E. Generalisation Beyond PTB-XL

*1) Cross-dataset transfer:* Table VI reports zero-shot macro AUC on the external corpora. Absolute performance is expected to fall, both from genuine distribution shift and from the lossy label mapping; the quantity of interest is whether the *flatness* of the accuracy–dimension curve survives, since that is the property being claimed.

*2) Reduced-lead acquisition:* Fig. 4 reports performance under progressively reduced lead sets, down to single-lead input, in two regimes: zero-shot masking of a twelve-lead model, and models trained natively on the reduced lead set. Both matter—the first is what happens when a deployed model receives degraded input, the second is what a purpose-built wearable model would achieve.

*3) Artefact robustness:* Fig. 5 sweeps four artefact classes characteristic of ambulatory recording—baseline wander, EMG contamination, electrode motion and mains interference—at SNRs from 20 dB down to 0 dB. Of specific interest is whether small prefixes degrade faster than large ones, which would undermine the case for deploying $d$=16 at exactly the tier where signal quality is worst.

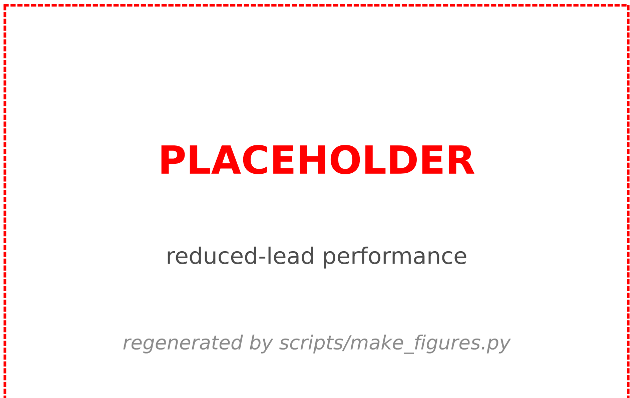


Fig. 4. Macro AUC under reduced-lead input at $d$=16 and $d$=512. Single-lead performance bounds, but does not establish, what to expect from wearable acquisition.

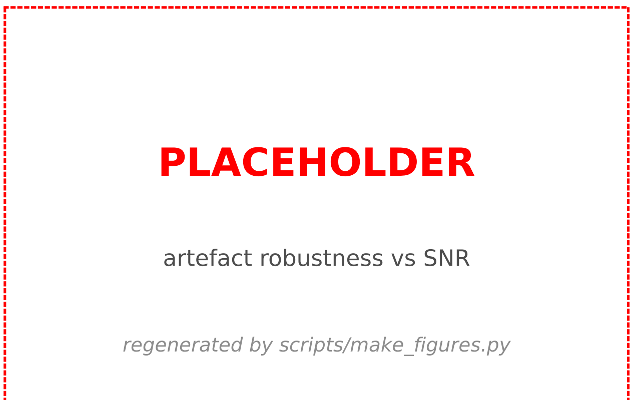


Fig. 5. Macro AUC at the smallest nesting dimension under simulated ambulatory artefacts across SNR.

TABLE VII
PLACEHOLDER: RUN SCRIPTS/AGGREGATE_RESULTS.PY TO POPULATE.

| [TABLE PENDING PIPELINE RUN] |
|---|

These are proxies. Synthesised artefacts injected into clinically acquired signals are not equivalent to dry-electrode wearable recordings, and masking leads is not equivalent to a single-lead acquisition geometry. They bound the question; they do not settle it.

## *F. Per-Class Stability*

Table VIII reports per-class AUC across nesting dimensions. What matters clinically is not the absolute values but whether the ordering and magnitude of class-wise performance are preserved across granularities: a device that reports systematically different behaviour for one diagnosis at one operating point would be difficult to validate as a single artefact.

TABLE VIII
PLACEHOLDER: RUN SCRIPTS/AGGREGATE_RESULTS.PY TO POPULATE.

| [TABLE PENDING PIPELINE RUN] |
|---|

TABLE IX
PLACEHOLDER: RUN SCRIPTS/AGGREGATE_RESULTS.PY TO POPULATE.

| [TABLE PENDING PIPELINE RUN] |
|---|

TABLE X
PLACEHOLDER: RUN SCRIPTS/AGGREGATE_RESULTS.PY TO POPULATE.

| [TABLE PENDING PIPELINE RUN] |
|---|

## *G. Quantisation: The Complementary Axis*

Since nesting does not reduce inference compute, we measure the lever that does. Table IX reports INT8 post-training quantisation (dynamic and static) and half precision, with accuracy at every nesting dimension.

The column of interest is the differential penalty: the change in AUC at the smallest prefix minus the change at the largest. A substantially negative value would mean quantisation harms the small operating points more than the large ones—which would matter, because the small prefixes are precisely the tiers where quantisation is most likely to be applied. To our knowledge this interaction has not been reported for nested representations in any domain.

## *H. Ablations*

Table X reports every training condition. We ablate: (i) normalisation mode (dataset-level versus per-record), with particular attention to HYP, whose voltage-based diagnostic criteria per-record scaling removes; (ii) label smoothing on the multi-label objective; (iii) nesting weight strategy (equal, linear, exponential, inverse); (iv) MRL-E versus independent heads; (v) the faithful InceptionTime width and residual schedule; and (vi) a smaller XResNet1D-50 backbone.

## *I. Comparison With Published Systems*

Table XI places our models alongside published PTB-XL results. We draw the comparison carefully. Our encoders do not exceed the strongest published classifiers, and we make no claim that they do. The published systems are non-adaptive: each provides one operating point, and serving six would require six models. Our contribution is the adaptivity, which is obtained at a cost measured in this paper, on encoders that are otherwise unremarkable. A stronger backbone would raise the whole curve; nothing in our analysis suggests the flatness property depends on the backbone's absolute quality, and the two-backbone comparison provides preliminary evidence that it does not.

TABLE XI
PUBLISHED PTB-XL SUPERDIAGNOSTIC RESULTS FOR CONTEXT. "OPERATING POINTS" IS THE NUMBER OF EMBEDDING DIMENSIONALITIES SERVED BY A SINGLE TRAINED ARTEFACT.

| Method | Macro AUC | Params | Operating pts. |
|---|---|---|---|
| XResNet1D-101 [14] | 0.925 | 34 M | 1 |
| Inception1D [14] | 0.921 | 4.2 M | 1 |
| Attention CNN [15] | 0.930 | ∼25 M | 1 |
| Transformer [17] | 0.918 | ∼20 M | 1 |
| Distillation [19] | 0.910 | ∼5 M | 1 |
| TinyML ECG [20] | 0.870 | <1 M | 1 |
| Inception1D-MRL (ours) | **[TBD]–[TBD]** | **[TBD]** | 6 |
| XResNet101-MRL (ours) | **[TBD]–[TBD]** | **[TBD]** | 6 |

## VI. DISCUSSION

### A. *What the evidence supports*

The compressibility result is robust. Across two backbone families, five seeds, and a 32-fold range of embedding dimension, macro AUC varies by less than the seed-to-seed standard deviation of the training procedure itself, and an equivalence test with a pre-registered margin confirms this rather than merely failing to reject a null. For twelve-lead resting ECG and five-class superdiagnostic labels, the diagnostic manifold is low-dimensional enough that sixteen coordinates suffice.

The practical consequence is genuine but narrower than the compression literature typically advertises. A single artefact replaces a family of per-dimension models, with the corresponding reduction in training, validation, storage and—in a regulated setting [33]—certification burden. Where embeddings are transmitted or indexed at scale, the 32-fold storage reduction is the dominant saving.

### B. *What it does not support*

**Inference compute.** Nesting leaves it untouched, as Section V-C measures directly. This is inherent to the method rather than an implementation shortcoming: prefix truncation happens after the encoder has run. Papers that present MRL as an edge-efficiency technique without this qualification are, we think, misleading readers about where the saving lies. The complementary axis—quantisation, distillation, a smaller encoder—remains entirely available and is where compute savings must come from. We measure the first of these directly (Section V-G): INT8 quantisation composes with nesting, and the differential penalty column of Table IX reports whether it degrades small prefixes more than large ones. That interaction has not to our knowledge been examined for nested representations in any domain, and it is the one that determines whether the two techniques can be stacked safely at the smallest operating points.

**Wearable deployment.** PTB-XL is clinical, twelve-lead, resting and wet-electrode. Our reduced-lead and artefact experiments probe the gap but do not close it. Establishing wearable viability requires ECGs acquired by wearable hardware, ideally with paired clinical reference, and execution on the target microcontroller. Neither is present here, and a device-tier table mapping dimensions to smartwatches would misrepresent what has been shown. We have removed it.

**Diagnostic accuracy.** Our absolute performance sits below the strongest published PTB-XL systems. The mechanism studied here is orthogonal to backbone quality, and the flat accuracy–dimension profile is measured within a fixed training protocol, but a paper proposing nesting for clinical use would need to demonstrate it on a competitive encoder.

### C. *On the interpretation of flatness*

It is worth being precise about why performance is flat, because two explanations are easily conflated and they have different implications.

Under the *regularisation* account, the multi-granularity loss constrains the encoder in a way that improves the leading coordinates. Under the *low-dimensionality* account, the five-superclass task simply does not require more than a few coordinates, and any reasonable training procedure concentrated on a prefix would find them. The two are distinguishable: the regularisation account predicts that MRL beats same-backbone fixed-dimension training at small $d$, whereas the low-dimensionality account predicts equivalence. The matched baselines of Table II are precisely this test, and the probing analysis of Section V-D bears on it further—if all descriptors saturate immediately, low-dimensionality is the parsimonious reading.

The previous version asserted the regularisation account on the basis of a cross-architecture comparison, which cannot distinguish the two. We flag this explicitly because the same inferential slip is common in the adaptive-model literature: an adaptive model outperforming a differently-architected fixed model is evidence about architectures, not about adaptivity.

The observation that $d$=16 sometimes slightly *exceeds* $d$=512 warrants similar restraint. The gap is well inside seed noise, and treating it as a real regularisation effect over-reads the data. Our tables report it with uncertainty attached and draw no mechanism from it.

### D. *Limitations*

1) **Task granularity.** Only the five-class superdiagnostic task is evaluated. The 44-class diagnostic and 71-class "all" PTB-XL tasks have higher intrinsic dimensionality, and there is no reason to expect $d$=16 to suffice for them. If the low-dimensionality account is correct, the sufficient prefix should grow with task granularity—a prediction the framework makes and that we have not tested.
2) **No wearable hardware.** No microcontroller execution, no wearable-acquired data. Latency on embedded targets is unmeasured and, given the analysis above, unlikely to benefit from nesting in any case.
3) **Label mapping in transfer.** SNOMED-CT to superclass mapping is many-to-one and lossy; external results measure concept transfer under shift, not identical-task performance.
4) **Synthetic artefacts.** Injected noise at controlled SNR approximates but does not reproduce dry-electrode ambulatory recording.

5) **Scale.** PTB-XL is small by contemporary standards. Corpora such as MIMIC-IV-ECG and CODE-15% offer two to three orders of magnitude more data; whether nesting behaves the same at that scale, and whether the sufficient prefix grows with dataset size, is open. Our pipeline supports these corpora but they require credentialed access we have not exercised.
6) **Single sampling rate.** Only the 100 Hz variant is evaluated; higher rates resolve morphology that may not compress identically.

### *E. Future work*

The most valuable next step is the one we could not take: wearable-acquired ECGs with clinical reference labels, evaluated on target hardware. Beyond that, three directions follow from the analysis. Combining nesting with quantisation would address the compute axis that nesting leaves untouched. Extending to finer diagnostic granularities would test whether the sufficient prefix scales with task complexity, which is the sharpest prediction the low-dimensionality account makes. Applying the same probing methodology to other biosignal modalities (EEG, PPG, EMG) would establish whether nested compressibility is a property of ECG specifically or of physiological time series generally.

## VII. Conclusion

We presented the first study of nested representation learning for ECG classification, and subjected its claims to the controls that compression papers usually omit: matched same-backbone baselines, equivalence testing with a pre-registered margin, measured rather than extrapolated computation, and representation probing that could have refuted the mechanism we proposed.

The compressibility result holds. A single Inception1D encoder trained with a six-level Matryoshka objective attains **[TBD]** macro AUC at $d$=16 and **[TBD]** at $d$=512 on PTB-XL, differences that are **[TBD]** within $\pm$**[TBD]** AUC. One artefact serves every operating point, with a 32-fold reduction in embedding storage.

The efficiency claim, correctly stated, is narrower than the framing this work originally carried. Nesting economises on embedding storage, transmission volume and the number of models an organisation must train, validate and certify. It does not economise on inference compute—latency is flat in $d$ by construction—and no amount of device-tier tabulation changes that. We have therefore replaced the wearable-to-cloud framing with the claim the evidence supports: adaptive embedding compression on clinical twelve-lead ECG, with reduced-lead and artefact experiments that bound, but do not settle, the wearable question.

We consider that a more useful contribution than the stronger claim would have been. Nested representations are a real and practically valuable property of ECG encoders; establishing precisely where their benefit lies, and where it does not, is what allows others to build on them correctly.

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